# PingSheng Kao

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## SUMMARY

ML Data Scientist with 5+ years of experience improving production vision, multimodal, LLM, and agentic systems through model evaluation, targeted fine-tuning, statistical analysis, and data pipelines. Turns failure patterns and quality metrics into actionable model improvements.

## EXPERIENCE

**ML Data Scientist - Apple** July 2020 - Present
Santa Clara, California

- Developed embedding-driven failure discovery for Vision Pro hand tracking using similarity search, clustering, OOD, and position-level analysis; contributed to a quality scorecard with final missing-pinch rate below 2% and false positives below 5 per hour.
- Trained and fine-tuned vision models on VLM-discovered and relighting-derived failure slices, benchmarking targeted model repair against alternative slice-discovery approaches.
- Built a simulated-user agent, agent-as-judge workflow, targeted scenario generators, and behavioral metrics to uncover visual-memory, sycophancy, intent-misalignment, and false inferred-state-update failures in an open-ended multimodal conversational agent.
- Built expert-aligned evaluation for a multimodal health-related AI product, including gold-set curation, expert/non-expert labeling workflows, and an auditable multi-stage LLM evaluator that reached 90% alignment with experts on a 40-example gold set.
- Produced leadership-facing model-release reports pairing scorecard metrics with regression attribution, embedding-cluster diagnosis, and online validation of offline-mined failure patterns.
- Wrote SQL in an evaluation pipeline querying AWS Athena to calculate model-quality metrics and export results to dashboards.

## RESEARCH CONTRIBUTION

**GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks** 2025
Experiment contributor (credited as Sam Kao) - accepted to ECCV - arXiv:2512.24592

- Contributed experiments using LLMs and VLMs to generate and verify grounded failure hypotheses for fine-grained, interpretable error-slice discovery.
- Trained and fine-tuned vision models on targeted failure slices and compared repair against alternative discovery approaches.

## ADDITIONAL EXPERIENCE

**Algorithm and Architecture Intern - Deep Learning Compression - Ambarella** June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

## EDUCATION

**M.S. in Computer Science and Engineering - University of Michigan, Ann Arbor** 2018 - 2020
**B.S. in Electronic Engineering - National Taiwan University** 2013 - 2017
GPA: 3.9/4.0 (B.S.)

## SKILLS

Programming: Python, SQL, C++, JavaScript
Modeling: PyTorch, Vision-Model Fine-Tuning, Neural-Network Quantization
Evaluation: Agent-as-Judge, LLM-as-Judge, Benchmark and Metric Design, Statistical Analysis, Human Evaluation
Data & Quality: AWS Athena, Embeddings, Clustering, OOD Analysis, Tableau, Failure Analysis